José Luis Rativa Medina

AI Engineer — Software Engineer

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PROFILE

AI Engineer and Software Engineer with hands-on experience building and deploying Generative AI and Agentic AI systems, including RAG, LLM distillation, conversational memory, and real-time voice agents. Strong software engineering background in TypeScript, React, Node.js, and Python, with research experience in generative models, computer vision, and IoT security, Experience working in agile environments and AI-augmented development workflows.

EDUCATION

Télécom SudParis – Institut Polytechnique de Paris Évry, France

Master's Degree in Engineering, Data and AI track 2024–2026

Relevant Coursework: Computer Vision, Deep Learning, Probability for Machine Learning, Reinforcement Learning, Large-scale Generative Models for NLP and Speech Processing, Relational and Distributed Data Management, Stochastic Processes, Ethics and Laws in AI and Data Science.

Universidad Nacional de Colombia Bogotá, Colombia

BSc in Systems and Computer Engineering 2020–2024

Relevant Coursework: Calculus, Linear Algebra, Software Engineering, Software Architecture, Introduction to Intelligent Systems, Object Oriented Programming, Data Structures, Compilers and Programming Languages, Algorithms, Project Management.

EXPERIENCE

ARY S.A.S Palaiseau, France

AI Engineer Intern March 2026 – September 2026

- Contributed to the end-to-end design and development of voice-based conversational agents, from architecture through deployment, integrating real-time speech recognition and synthesis to enable natural, low-latency voice interactions.
- Designed and implemented RAG architectures and conversational memory components, enabling agents to leverage clients' knowledge bases and maintain contextual continuity across long, multi-turn interactions.
- Applied knowledge distillation from Mistral Large to a Qwen 0.5B student model for user-intent classification, followed by quantization and WebLLM/WebGPU optimization to enable lightweight, client-side inference.
- Developed a lightweight ONNX lip-sync model through knowledge distillation from MuseTalk, using a compact convolutional architecture and replacing MuseTalk's VAE with TAESD to reduce memory and computational requirements for WebGPU deployment.

Universidad Nacional de Colombia Remote

Software Engineer November 2023 – August 2026

- Contributed to the architecture and development of a web platform for student pre-registration for Colombia's national Saber Pro exam (Icfes), supporting a reliable and scalable academic assessment workflow.
- Contributed to the architecture and design of a self-assessment system for undergraduate and graduate programs, enabling the evaluation and monitoring of academic programs.
- Designed database schemas and performed ongoing maintenance of information systems, ensuring data integrity, reliability, and operational continuity.

RELEVANT ACADEMIC PROJECTS

- **3D Object Reconstruction via Diffusion Models** *(Python, PyTorch)* – Applied generative diffusion models to reconstruct 3D objects from 2D slices.
- **Privacy-Preserving Identity for IoT** *(Hyperledger Fabric, Blockchain, IoT)* – Designed and implemented an end-to-end decentralized identity and access-control architecture for IoT environments.
- **TiendaUn** *(Docker, RabbitMQ, REST/GraphQL, React, Express.js)* – Designed and implemented an event-driven microservices architecture for a university e-commerce platform.
- **3D Human Segmentation** *(PyTorch)* – Developed deep learning methods for human silhouette segmentation in 3D scenes.

SKILLS

Programming: Python, TypeScript, JavaScript, Java
OS:Ubuntu, Omarchy, Arch Linux
AI Development: Claude Code, OpenCode, AI-augmented development workflows
AI/ML: PyTorch, Hugging Face Transformers, LangChain, ONNX
Web: React, Node.js, Express.js, FastAPI, Django
Data: PostgreSQL, Supabase, MySQL, MongoDB, pgvector, Pinecone
Cloud & Tools: AWS, GCP, Docker, Git, Jira
Languages: Spanish (Native), English (C1), French (C1)
Certification: AI Agent Developer, Coursera
Soft Skills: Problem solving, analytical thinking, effective communication, teamwork and adaptability.